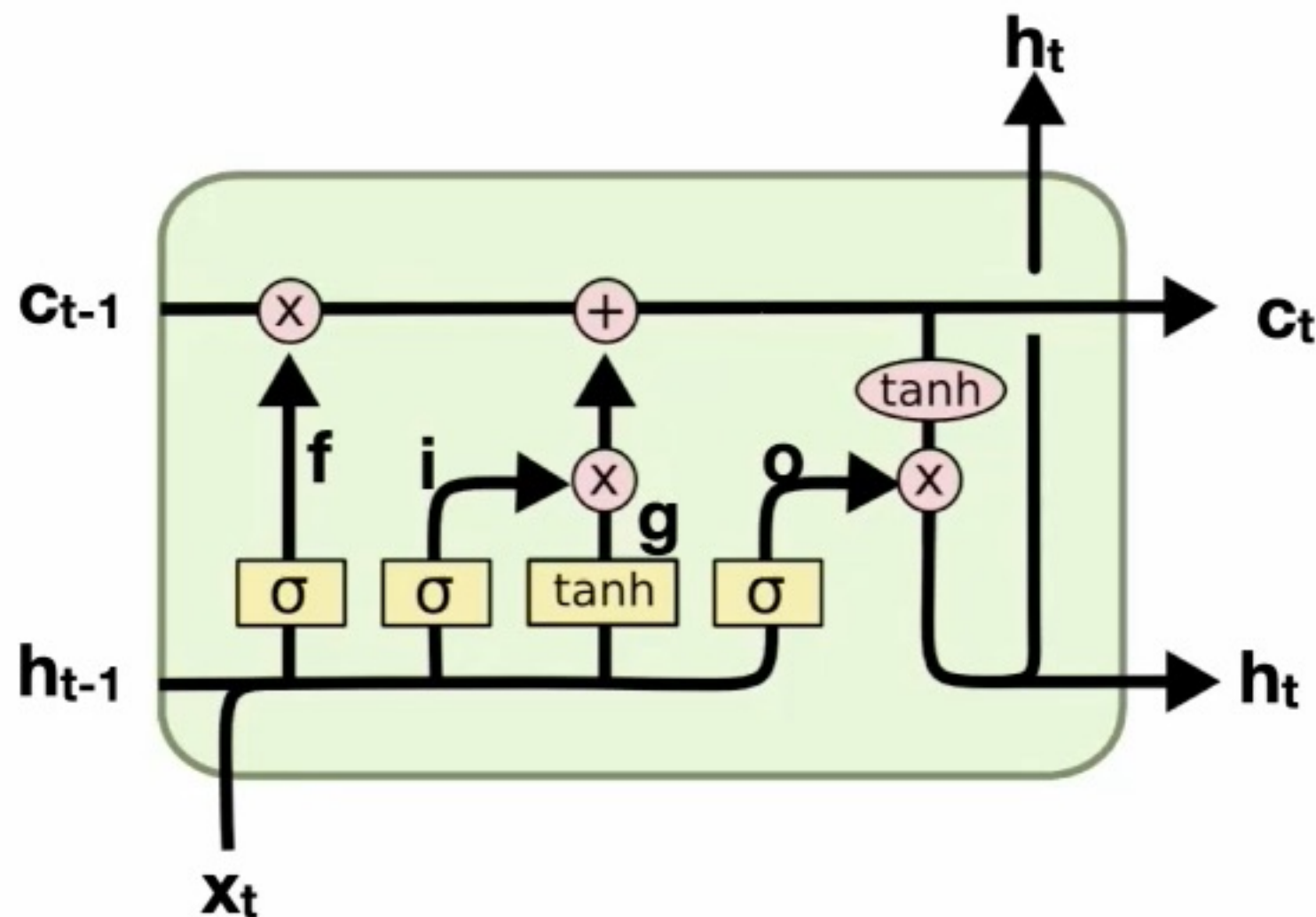


LSTM

Long Short-Term Memory [Hochreiter & Schmidhuber, 1997]

- A popular option for the function **R**
- Split state vector into two halves: "memory cell" **c** and working memory **h**.
- "gates" decide:
 - how much of the input to add to the memory
 - how much of current memory to forget



f: forget gate - erase memory cell?

i: input gate - write input to memory cell?

g: (?) gate - how much input to write to memory cell.

o: output gate - how much of the cell should be used for the output.